Foreground–Background Contrastive Learning for Few-Shot Remote Sensing Image Scene Classification

Jie Geng, *Member, IEEE*, Bohan Xue, and Wen Jiang

***Abstract*— Few-shot learning (FSL) aims to train a model with limited samples for identifying novel category samples. As for remote sensing (RS) images, complex backgrounds may lead to large intraclass differences, and the number of labeled samples is quite smaller than that of large datasets, which both influence the classification performance. To solve these issues, a foreground–background contrastive learning (FBCL) is proposed for few-shot RS image scene classification. Specifically, a foreground–background separation (FBS) module is proposed to separate features between objects and background with supervised contrastive learning (SCL), which aims to improve the ability to distinguish foreground and background regions of RS images. Moreover, a channel weight allocator is proposed to balance features of different dimensions, which can take full advantage of RS image information. Experiments on three RS datasets prove that the proposed few-shot method is able to produce superior classification results than other related approaches.**

***Index Terms*— Contrastive learning, few-shot learning (FSL), remote sensing (RS) image, scene classification.**

I. Introduction

With the progress of imaging technology, more remote sensing (RS) data with higher resolutions are captured for various applications. RS scene classification is a basic processing technique for RS interpretations, which has been adopted in urban management [1], disaster observation [2], target monitoring [3], geological survey [4], and so on. Recently, deep neural networks (DNN) have been applied in numerous fields successfully. Different convolutional neural networks (CNNs), including VGG [5], GoogleNet [6], and ResNet [7], have been applied for scene classification.

Although deep-network-based methods achieve superb performance for scene classification, they generally have complex structures and rely on extensive labeled training samples [8], [9]. As for various practical scenarios, it is quite difficult to annotate numerous data, due to factors such as privacy, security, or high labeling cost [10], [11]. Without enough labeled training samples, overfitting generally occurred during the training of DNN [12], [13]. To reduce the overfitting, Ghazi et al. [14] developed data enhancement methods, where image transformations including translation, rotation, and scaling are used. Srivastava et al. [15] proposed dropout that randomly removes the units from the deep network to prevent excessive co-adaptation among the units. Although data enhancement and regularization techniques can partially reduce the shortage of training samples, the performances in various tasks are not satisfactory.

Most recently, few-shot learning (FSL) has been proposed to train a deep network with limited samples to acquire the ability of identifying novel category samples. It aims at quickly extracting information from just fewer labeled samples to predict unknown samples. Various meta-learning-based FSL models have been proposed [16], [17], where meta-learning-based methods divide the few-shot classification task into a series of subtasks. Meta-learning uses episode training to expect model learning to learn, which owns the ability to rapidly adjust to new tasks. Therefore, meta-learning models have been developed for RS scene classification. Li et al. [18] proposed a fast adaptation model, which applies the self-supervised embedding optimization for adaptive generalization in new tasks. Gong et al. [19] combined meta-learning with self-supervised learning to adapt the model to the target domain.

The above-mentioned methods mainly research a novel measurement between the training and test samples, which ignore the domain gaps between the training and test samples [20], [21]. It may weaken the effectiveness of meta-learning. Zhang et al. [22] introduced graph information aggregation into few-shot tasks to achieve domain adaptation between the source data and target data. Tian et al. [23] argued that extracting extensive and more representative features from the samples is the key to resolve the domain gap issue. Thus, instead of using meta-learning, they trained the feature extractor by integrating the labeled samples of all the subtasks, which yields superior performance than other meta-learning approaches. It can be observed that a well-performed feature extractor is of great importance for meta-learning.

As for RS images, there are larger intraclass variability and smaller interclass variability in some cases. It is clear from Fig. 1 that different categories of objects may appear in the same scene with diverse scales and orientations, while the same category objects may be presented in different scenes.

Manuscript received 27 March 2023; revised 20 May 2023; accepted 22 June 2023. Date of publication 28 June 2023; date of current version 10 July 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62271396, in part by the Shaanxi Key Research and Development Program under Grant 2023-YBGY-220 and Grant 2022ZDLGY03-04, in part by the Project funded by China Postdoctoral Science Foundation under Grant 2021TQ0271 and Grant 2021M700110, and sponsored by the Practice and Innovation Funds for Graduate Students of Northwestern Polytechnical University under Grant PF2023014. *(Corresponding author: Jie Geng.)*

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Digital Object Identifier 10.1109/TGRS.2023.3290794

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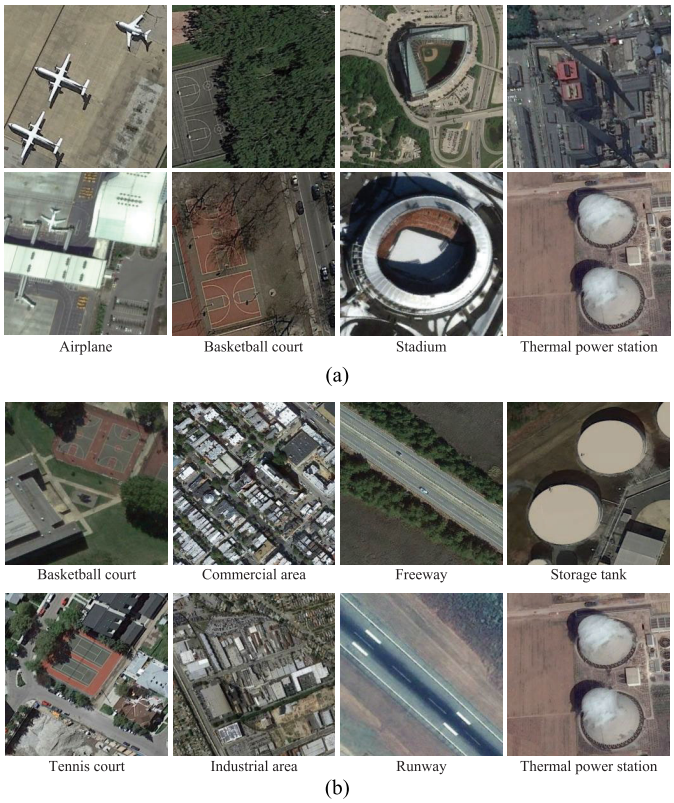


Fig. 1. Images selected from NWPU-RESISC45. (a) Large intraclass variability. (b) Small interclass variability.

To resolve this issue, Zeng et al. [24] proposed a model with the self-attention property to diminish the interference of irrelevant information. Kim and Chi [25] proposed a multiscale fusion network for RS images to extract more valuable texture features. Although these methods can extract rich semantic information from RS images, it cannot distinguish the foreground and background semantics effectively.

To address the above issues, a foreground–background contrastive learning (FBCL) is proposed for scene classification of RS images. In our proposed framework, supervised contrastive learning (SCL) is developed to maximize the consistency of intraclass samples while minimizing the similarity of interclass samples. To achieve this goal, a multilayer feature fusion (MFF) module is used for fusing features of different scales. Subsequently, a feature separation mask is trained to obtain the foreground and background features of each sample. At the same time, a novel contrastive loss is designed to increase the foreground similarity of intraclass samples and to reduce the correlation between foreground and background features. Moreover, a multifeature channel weight allocator is developed to stitch the foreground and background features according to the channel dimension. In brief, this article's main contributions are summarized below.

1) In our framework, a foreground–background separation (FBS) model is proposed with supervised contrast learning to yield feature separation masks. It is able to effectively distinguish the foreground and background regions of the images.
2) An MFF module is proposed during feature learning, where shallow features and deep features from the backbone network are fused by the mutual attention mechanism. It is able to effectively capture global information and enhance the discrimination of feature representation.
3) A multifeature channel weight allocation (MCWA) is proposed to stitch the foreground and background features in the channel dimension by optimizing the weights of features. It can take full advantage of features learned from RS images, which contributes to achieve better accuracy of few-shot classification.

The rest of this article is organized as follows. Section II presents related work of scene classification, FSL, and contrastive learning. The proposed FBCL is introduced in Section III. In Section IV, the experimental results are presented and analyzed. Finally, conclusions are concisely summarized in Section V.

II. Related Work

A. RS Image Scene Classification

Scene classification of RS image is to classify images into different categories based on scene information [26], [27]. Although RS images capture more features including color, texture, and shape, it faces problems of complex contexts and different receptive field targets. Therefore, scene classification of RS data faces a greater challenge than natural image classification [28].

The traditional approaches for scene classification of RS data are combining the traditional classifiers with low level and middle level features. Van De Sande et al. [29] developed a classification method for modeling visual scenes on the basis of local invariant features and probabilistic latent space model. Feng et al. [30] used bag-of-visual-words (BOV) and clonal selection model to extract more appropriate visual words from the underlying features, which can effectively promote the classification effects. Recently, deep learning has advanced rapidly in many domains, which has also promoted the performance improvement of RS classification. Yu et al. [31] proposed an end-to-end feature fusion model to aggregate features from dual paths. Zhang et al. [32] developed a network for semantic message and topological structure passing, which uses graph structure to describe the topological relationship between features. These mentioned approaches generally require a large number of labeled data for training, which is hard to satisfy in many practical applications.

In the case of insufficient training samples, semi-supervised learning (SSL) is a popular solution, which aims to select pseudosamples by digging the data distribution of the unlabeled samples, thus expanding the training set. Semi-supervised classification can be mainly categorized into transductive methods and inductive methods, which has been developed for scene classification. Tang et al. [8] proposed a class-level prototype-guided multiscale feature learning model, which can effectively obtain high confidence unlabeled samples as pseudosamples. Han et al. [33] proposed a generative framework of SSL for collaborative training, which can excavate

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useful information from unlabeled data and expand the training set. Semi-supervised methods can reduce the influence of insufficient training samples, but they generally require a great number of iterations and depend on the quality of pseudosamples.

B. Few-Shot Learning

FSL aims to train a deep network with insufficient labeled samples. In many specific fields, such as military, medicine, and economics, it is quite hard to obtain adequate labeled samples. FSL approaches have a significant advantage to learn from fewer labeled samples, which have received increasing attention [34].

There are three main approaches for FSL, which are fine-tuning-based methods, data-augmentation-based methods, and transfer-learning-based methods. The fine-tuning models are first trained with a great number of labeled samples on the source domain, and then fine-tuned on the target domain which contains fewer labeled samples. This approach generally leads to overfitting when samples of different domains reveal large gaps. The data–augmentation-based methods extend the target domain data with auxiliary information, which can reduce the overfitting issues caused by insufficient samples. The transfer-learning-based methods aim to transfer learned knowledge crossing different domains, which mainly include methods on the basis of metric learning and meta-learning.

To realize few-shot classification, the metric-learning-based models calculate a certain distance between labeled and unlabeled samples, where the distance is applied as the basis for category predictions. Vinyals et al. [17] proposed a matching network that uses long short-term memory for information transfer. Snell et al. [35] proposed a prototype network for FSL, which learns prototypes to represent class centers. Sung et al. [36] proposed a relational network, which analyses the matching degree between unlabeled samples and the support set by constructing a neural network. Xie et al. [37] proposed a few-shot classification model based on depth Brownian distance covariance, which is able to obtain more efficient similarity by measuring the joint distribution of sample pairs. Wertheimer et al. [38] reconstructed the query set features by learning the transformation matrix between the support set and the query set, where the similarity between sample pairs is also used for few-shot classification.

The meta-learning-based approaches have been proposed for FSL, which can train the deep model to acquire the capacity of learning to learn. With this capacity, meta-learning models can learn novel knowledge quickly from fewer samples based on the prior knowledge. Lee et al. [39] proposed the MetaOptNet, which trains linear predictors differentially for different subtasks to improve the generalization. Hsu et al. [40] proposed an unsupervised meta-learning approach, which constructs classification tasks from unlabeled data in an automatic manner, so that the model is able to excavate useful prior knowledge for downstream classification tasks.

C. Contrastive Learning

Contrastive learning focuses on learning consistent features of the same categories and distinguishing characteristics among different category samples. Generally, contrastive learning approaches can be split into self-SCL and SCL.

As for self-SCL, each sample is randomized twice to form the positive sample pairs, which expects the model to decrease the distance of positive samples and increase the distance of negative samples in the feature space as much as possible. He et al. [41] proposed the momentum contrastive (MOCO) model, where contrastive learning is considered as a dynamic dictionary to conduct a large and consistent unsupervised learning. The MOCO model yields effective results with simple linear protocol for ImageNet classification. Chen et al. [42] proposed a simple contrastive learning framework (SimCLR), which is demonstrated to be suitable for data enhancement. In the SimCLR model, the feature representations are passed through a learnable nonlinear transformation module to enhance the quality of the features. Therefore, it is able to learn more discriminative features from images using self-SCL, and the transfer ability of the deep model can be improved at the same time [43]. However, self-SCL cannot take advantage of the associations among same category samples.

SCL organizes positive instances by the same category samples and negative instances by different category samples, which aims to learn more representative category features. Khosla et al. [44] effectively used label information to aggregate the same category samples together in the embedding space and determine the optimal loss formula. Zheng et al. [45] adopted the weakly labeled samples to conduct SCL, which is able to enhance the classification accuracy across datasets. Thus, SCL tends to yield superior performance than self-supervised learning methods.

Inspired by this, SCL is applied as an auxiliary task in our work, where an FBCL model is proposed for RS image scene classification. Different from the above methods, our proposed FBCL developed a novel SCL between the foreground and background, which aims to efficiently distinguish the foreground and background regions of the images. It can take full advantage of features learned from RS images. Thus, different from typical contrastive learning that constructing positive and negative sample pairs for data enhancement, our contrastive learning model learns foreground and background features by constructing positive and negative sample pairs of foreground–background.

III. Methodology

A. Overview

In this work, an FBCL is proposed for scene classification of RS images. The proposed framework is composed of an MFF model, an FBS model, and an MCWA. Specifically, the MFF model is applied for fusing features under different scales from the backbone model. The FBS model based on SCL is proposed, which can maximize the consistency of intraclass samples and minimize the similarity of interclass samples. The multifeature channel weight allocation model is developed to stitch the foreground and background features according to the channel dimension, which can take full advantage of

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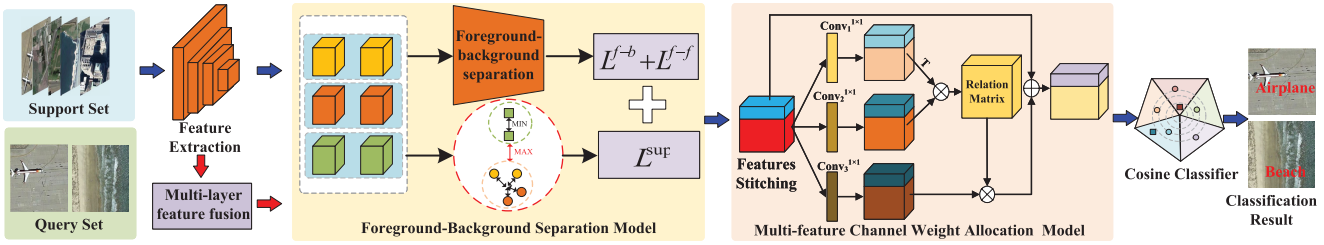


Fig. 2. Framework of the proposed FBCL for few-shot RS image scene classification. The model includes two stages: In the first stage, an MFF module and an FBS module based on SCL are constructed for training; in the second stage, a multifeature channel weight allocation module is proposed for meta-training.

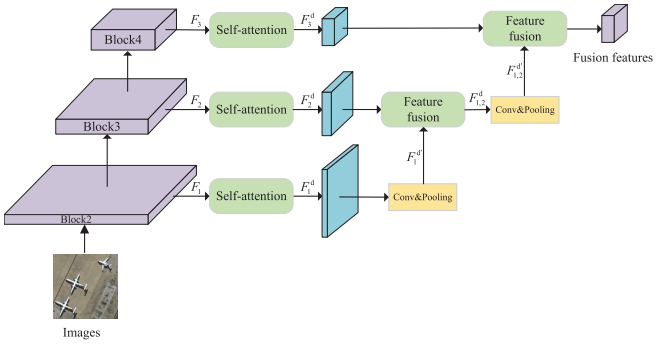


Fig. 3. Structure of the MFF module.

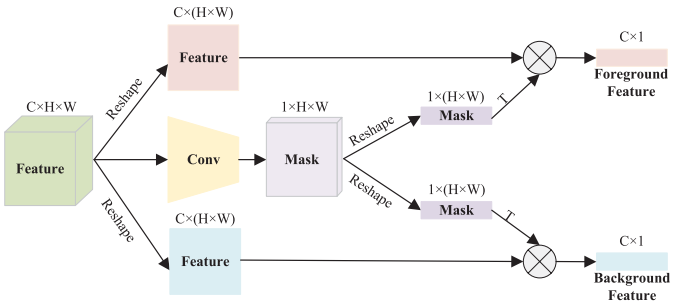


Fig. 4. Structure of the FBS module.

features learned from RS images. The overall framework of the proposed FBCL is shown in Fig. 2.

As for the classification task, the dataset is generally divided into three parts: training data D_{train}, validation data D_{val}, and test data D_{test}. To conduct FSL, the training data can be split into a support set D_s and a query set D_q. For each episode of training, randomly select N categories from the training data, where K samples of each category are chosen as the support set and Q samples as the query set.

B. Multilayer Feature Fusion

CNN abstracts image features based on sliding windows, which has preferable ability of capturing local features. As for feature extraction of RS images with complex semantics, with the deepening of CNN, it results in the loss of features during pooling. To obtain a more comprehensive feature representation, we develop an MFF module, which extracts global representations of single-layer features through the self-attention module [46] and mines correlations between different scales of features through step-by-step fusion.

In this work, ResNet-12 is used as the backbone, which consists of four residual blocks and a global averaging pooling layer. Shallow features have a low degree of abstraction and contain low-level image information, while deep features are highly abstract and contain high-level semantic information. To represent features of the RS image more comprehensively, features $\mathbf{F}_1 \in \mathbb{R}^{C_1\times H_1\times W_1}$, $\mathbf{F}_2 \in \mathbb{R}^{C_2\times H_2\times W_2}$, and $\mathbf{F}_3 \in \mathbb{R}^{C_3\times H_3\times W_3}$ of the last three residual blocks are imported into the self-attention module, respectively, where H_i and W_i denote the length and width of the feature, respectively, and C_i stands for the feature dimension. This process can be expressed as

$$\mathbf{F}_i^d = \text{SelfAttention}(\mathbf{F}_i), \quad i \in \{1, 2, 3\}. \tag{1}$$

After that, different layer features are fused in a step-by-step process. As shown in Fig. 3, features $\mathbf{F}_1^d$ and $\mathbf{F}_2^d$ are fused to obtain feature $\mathbf{F}_{1,2}^d$, and then $\mathbf{F}_{1,2}^d$ and $\mathbf{F}_3^d$ are fused to obtain the final output feature $\mathbf{F}^d$. To conduct fusion, we adjust the channel of $\mathbf{F}_1^d$ to C_2 by a 1×1 convolution layer and the space size to H_2 and W_2 by an average pooling layer, which can be expressed as: $\mathbf{F}_1^{d'} = \text{AvgPool}(\text{Conv}^{1\times1}(\mathbf{F}_1^d))$. Then, mapping feature $\mathbf{F}_2^d$ to Q and feature $\mathbf{F}_1^{d'}$ to K and V, and calculating the correlation coefficient between Q and K through dot product to update Q. The above process can be expressed as follows:

$$\mathbf{Q} = \mathbf{W}_q \cdot \mathbf{F}_2^d, \mathbf{K} = \mathbf{W}_k \cdot \mathbf{F}_1^{d'}, \mathbf{V} = \mathbf{W}_v \cdot \mathbf{F}_1^{d'} \tag{2}$$

$$\mathbf{F}_{1,2}^d = \mathbf{Q} + \text{Softmax}\left(\frac{\mathbf{Q}\cdot\mathbf{K}^T}{\sqrt{d_k^h}}\right)\cdot\mathbf{V} \tag{3}$$

where $\mathbf{W}_q$, $\mathbf{W}_k$, and $\mathbf{W}_v$ are learnable linear transformations. The fusion process between $\mathbf{F}_{1,2}^d$ and $\mathbf{F}_3^d$ is the same as above, so as to obtain the final fused feature $\mathbf{F}^d \in \mathbb{R}^{C_3\times H_3\times W_3}$.

C. Foreground–Background Separation

As for RS images, there are issues of large intraclass variations and small interclass differences due to the complex background. At the same time, objects in RS images have the characteristics of diversity and multiscales, which are hard to discriminate from the background in some cases. To resolve these problems, an FBS module with SCL is proposed to effectively distinguish the foreground and background regions of RS images, which aims to reduce the interference of background and improve the quality of learned features. The structure of the FBS module is shown in Fig. 4.

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After the MFF module, which is denoted as $f(\cdot)$, we can obtain features $\mathbf{F}^d$ and $\mathbf{F}_3$. Here, $\mathbf{F}^d$ represents the fused features and $\mathbf{F}_3$ represents the last layer features in the backbone before the global averaging pooling layer. Through the FBS module, the mask $\mathbf{M}$ is learned to distinguish the foreground and background, whose values corresponding to each pixel are ranging from 0 to 1. Then, the masks $\mathbf{M}$ can be multiplied with the original features $\mathbf{F}^d$ to obtain the corresponding foreground features. The process of $\mathbf{F}^d$ can be expressed as below

$$\mathbf{M}_{\mathrm{I}} = m(\mathbf{F}^d; \theta_{\mathrm{I}}) \tag{4}$$

$$\mathbf{F}_{\mathrm{I}}^f = \mathbf{F}^d \otimes \mathbf{M}_{\mathrm{I}} \tag{5}$$

$$\mathbf{F}_{\mathrm{I}}^b = \mathbf{F}^d \otimes (\mathbf{1} - \mathbf{M}_{\mathrm{I}}) \tag{6}$$

where $m(\cdot)$ stands for the FBS model, and θ_{I} denotes the corresponding parameter. $\mathbf{F}_{\mathrm{I}}^f \in \mathbb{R}^{C_3 \times 1}$ and $\mathbf{F}_{\mathrm{I}}^b \in \mathbb{R}^{C_3 \times 1}$ represent the foreground feature and background feature separating from $\mathbf{F}^d$ after masking, respectively. In the same way, we can obtain the foreground feature $\mathbf{F}_{\mathrm{II}}^f$ and background feature $\mathbf{F}_{\mathrm{II}}^b$ separating from the feature $\mathbf{F}_3$. Afterward, we fuse these features to obtain the foreground features $\mathbf{F}^f$ and background features $\mathbf{F}^b$ by averaging, i.e., $\mathbf{F}^f = (\mathbf{F}_{\mathrm{I}}^f + \mathbf{F}_{\mathrm{II}}^f)/2$ and $\mathbf{F}^b = (\mathbf{F}_{\mathrm{I}}^b + \mathbf{F}_{\mathrm{II}}^b)/2$.

To train the FBS model, a novel loss function measuring foreground–background and foreground–foreground information is proposed. To improve the ability of distinguishing foreground and background regions of RS images, the foreground and background features should be differentiable efficiently. The foreground features of the same category should be similar due to consistent characters. Therefore, our loss functions can be defined as follows:

$$\mathcal{L}^{f-b} = -\frac{1}{4n^2} \sum_{i=1}^{2n} \sum_{j=1}^{2n} \log\left(R\left(\mathbf{F}_i^b, \mathbf{F}_j^f\right)\right) \tag{7}$$

$$R\left(\mathbf{F}_i^b, \mathbf{F}_j^f\right) = \begin{cases} 1, & \text{if } \cos\left(\mathbf{F}_i^b, \mathbf{F}_j^f\right) < \rho \\ 1 - \cos\left(\mathbf{F}_i^b, \mathbf{F}_j^f\right), & \text{otherwise} \end{cases} \tag{8}$$

$$\mathcal{L}^{f-f} = -\frac{1}{2n} \sum_{i=1}^{2n} \frac{1}{2n_{y_j} - 1} \sum_{j=1}^{2n} \mathbb{I}_{[i \neq j]} \cdot \mathbb{I}_{[y_i = y_j]} \log\left(\cos\left(\mathbf{F}_i^f, \mathbf{F}_j^f\right)\right) \tag{9}$$

where $\mathcal{L}^{f-b}$ stands for the loss between the foreground and background features, $\mathcal{L}^{f-f}$ stands for the loss between the foreground and foreground features, and n represents the quantity of data in each batch. $R(\mathbf{F}_i^b, \mathbf{F}_j^f)$ indicates the similarity measurement between the background features $\mathbf{F}_i^b$ and foreground features $\mathbf{F}_j^f$, which is defined in (8). $\cos(\cdot)$ denotes the cosine similarity, and ρ denotes the threshold parameter. In our contrastive learning, each RS image is randomly processed twice, and thus it can obtain $2n$ foreground features and $2n$ background features in total. y_i stands for the label of the ith sample, and n_{y_j} represents the sample number belonging to the category y_j. $\mathbb{I}_{[\cdot]}$ is equal to 1 if the condition is satisfied; otherwise, it equal to 0. The loss function in (7) aims to increase the distance of negative sample pairs in feature space, that is to say, enlarging the discrepancy between the foreground feature and background feature. When the distance between them is large enough, it can be regarded that the model can easily distinguish this negative sample pair. To simplify the model training, negative sample pairs with larger distances are restricted by the proposed loss function, so the threshold ρ is set in (8). Thus, it can be seen from (8) that when the similarity between negative sample pairs is less than the threshold ρ, the loss is equal to 0.

To make our FBS module learn more discriminative category features, SCL loss is used in our work. In the supervised contrastive loss, the same category samples are set as positive pairs, and samples of different categories are set as negative pairs. Before this, the final layer feature $\mathbf{F}_3$ of the model is passed through the global average pooling layer and the projection head $g(\cdot)$. This process is expressed as follows:

$$\mathbf{G} = g(\mathrm{GlobalAvgPool}(\mathbf{F}_3)) \tag{10}$$

where $g(\cdot)$ consists of two linear layers and an ReLU layer. The loss function for SCL can be written as below

$$\mathcal{L}^{\mathrm{sup}} = -\sum_{i=1}^{2n} \frac{1}{2n_{y_j} - 1} \sum_{j=1}^{2n} \mathbb{I}_{[i \neq j]} \cdot \mathbb{I}_{[y_i = y_j]} \log \frac{\exp(\mathbf{G}_i \cdot \mathbf{G}_j)/\tau}{\sum_{k=1}^{2n} \mathbb{I}_{[i \neq k]} \exp(\mathbf{G}_i \cdot \mathbf{G}_k/\tau)} \tag{11}$$

where $\mathbf{G}_i$ and $\mathbf{G}_j$ denote the contrasting features corresponding to the ith and jth sample, respectively, and τ is the temperature coefficient. Finally, the total loss function to optimize the FBS model can be organized as follows:

$$\mathcal{L} = \alpha(\mathcal{L}^{f-b} + \mathcal{L}^{f-f}) + \mathcal{L}^{\mathrm{sup}} \tag{12}$$

where α is the scale factor to balance the proposed loss and the supervised contrastive loss.

D. Multifeature Channel Weight Allocation

As for FSL, different backgrounds of same category images and similar backgrounds of different categories are interferences for RS scene classification. On the other hand, other cases such as similar backgrounds of the same category images have a contribution to improve the performance of scene classification. Therefore, background information should be reasonably used for few-shot classification. To address this issue, multifeature channel weight allocation model is proposed to stitch the foreground and background features in the channel dimension. To realize channel weight allocation, the overall features of the image are used to weight the foreground channels and background channels.

The structure of the multifeature channel weight allocation module is shown in Fig. 5, which takes five-way one-shot as an example. The foreground features $\mathbf{F}_s^f$ and background features $\mathbf{F}_s^b$ of the support set D_s are obtained by the FBS module, whose overall dimensions are $N \times K \times C$. Similarly, foreground features $\mathbf{F}_q^f$ and background features $\mathbf{F}_q^b$ of the query set D_q with overall dimensions $N \times Q \times C$ can be obtained. As for $\mathbf{F}_s^f$ and $\mathbf{F}_s^b$, the corresponding foreground prototypes and background prototypes can be calculated by the mean value of features with the size of $N \times C$, i.e.,

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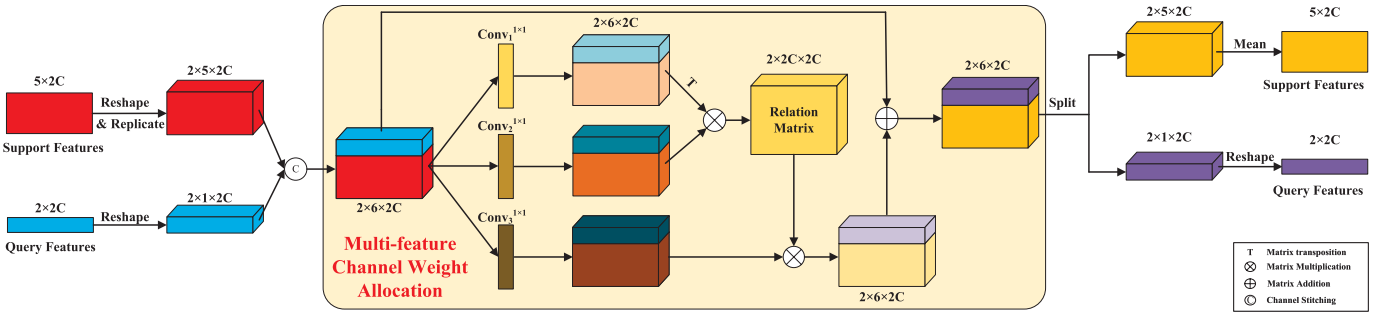


Fig. 5. Structure of the multifeature channel weight allocation module.

$\mathbf{P}^f = (1/K)\sum_{i=1}^{K} \mathbf{F}_{s_i}^f$, and $\mathbf{P}^b = (1/K)\sum_{i=1}^{K} \mathbf{F}_{s_i}^b$. Then, the prototypes of the foreground and background are stitched together to yield $\overline{\mathbf{P}} \in \mathbb{R}^{NQ\times N\times 2C}$, which are concatenated on the third dimension and replicated $Q-1$ times on the first dimension. At the same time, $\mathbf{F}_q^f$ and $\mathbf{F}_q^b$ are stitched similar to obtain features $\mathbf{F}_q \in \mathbb{R}^{NQ\times 1\times 2C}$ of D_q. After that, the prototype $\overline{\mathbf{P}}$ and feature of the query set $\mathbf{F}_q$ are spliced on the second dimension to yield $\mathbf{Z} \in \mathbb{R}^{NQ\times(N+1)\times C}$. $\mathbf{Z}$ is used to excavate the relational weight between $\overline{\mathbf{P}}$ and $\mathbf{F}_q$, which can be defined as follows:

$$\mathbf{A} = \sigma\left(\frac{\left(\text{Conv}_1^{(k\times k)}(\mathbf{Z})\right)^T \cdot \text{Conv}_2^{(k\times k)}(\mathbf{Z})}{\sqrt{C}}\right) \tag{13}$$

where $\sigma(\cdot)$ denotes the softmax function, C is the feature dimension, and $\sqrt{C}$ stands for the scaling factor to enhance the generalization ability. $\text{Conv}_i^{(k\times k)}$ denotes the convolutional layer with the kernel of $k \times k$. On the basis of relational weight $\mathbf{A}$, the allocated feature $\mathbf{Z}'$ can be calculated. The allocated prototype of the support set $\mathbf{P}'$ and allocated feature of the query set $\mathbf{F}'_q$ are obtained by splitting $\mathbf{Z}'$. The process function can be expressed as

$$\mathbf{Z}' = \mathbf{Z} + \text{Conv}_3^{k\times k}(\mathbf{Z}) \cdot \mathbf{A}^T \tag{14}$$

$$[\mathbf{P}', \mathbf{F}'_q] = \text{Split}(\mathbf{Z}') \tag{15}$$

where $\text{Split}(\cdot)$ stands for the splitting operation of the matrix. Finally, calculate the cosine similarity between the allocated feature of the query set $\mathbf{F}'_q$ and the allocated prototype of the support set $\mathbf{P}'$, which can predict the labels of the query set according to the highest similarity.

IV. Experiments

This section first describes the experimental datasets and the parameter settings. Then, the experimental comparisons of the proposed model and related FSL methods are reported. Finally, ablation experiments and visual analyses are conducted to further prove the efficacy of the proposed model.

A. Dataset

In this article, three public RS datasets are used in the experiments, which are Northwestern Polytechnic University remote sensing image dataset (NWPU-RESISC45) [47], UC Merced land use dataset (UCM) [30], and aerial image dataset (AID). We follow the experimental setup in [48] for the specific divisions for training, validation, and test.

B. Parameter Settings

In our proposed model, ResNet-12 is used as the backbone for feature extraction. For our contrastive learning, the setup is the same as SimCLR, where each image is randomly processed twice through data enhancement operations such as random cropping, horizontal rotation, and adjustment of image attributes.

During training the MFF and FBS modules, some empirical parameters followed the SCL [44]. Thus, the weight decay coefficient in the stochastic gradient descent (SGD) optimizer is set to 0.0001, and the learning rate is set to 0.5. The model is trained for 1000 epochs, and the learning rate decays at the 500th, 600th, 700th, and 800th epochs, respectively. In our proposed loss function, hyperparameters α, ρ, and τ are set to 1.8, 0.01, and 10, respectively.

During training the MCWA module, the empirical parameters followed meta-baseline [49], where the initial learning rate and weight decay coefficient are set to 0.001 and 0.0005, respectively. We used the PyTorch framework and two NVIDIA GeForce RTX 2080Ti GPUs for our experiments.

C. Experimental Comparisons

To prove the effectiveness of the proposed FBCL, several few-shot classification methods are adopted for comparisons, including matching network [19], prototype network [35], relational network [36], meta-baseline [49], DLA-MatchNet [50], SCL-MLNet [48], TSC [51], SPNet [52], CPT [53], AO [54], IDLN [55], and SGMNet [56]. For fair comparisons, we select the experimental results of AO model without transductive learning for comparisons. In addition, the AMP [57] algorithm is introduced in the CPT model, so we also conduct the experimental results with AMP. Performances of all the few-shot methods are evaluated in both the five-way one-shot and five-way five-shot cases, where the average accuracy of each method is reported on the test set.

Table I shows the compared results on the NWPU-RESISC45 dataset. It can be observed that compared with the classical FSL method, our proposed method has achieved a great improvement. Compared with the most recent works, the proposed FBCL still achieves superior classification accuracies. The proposed FBCL yields higher few-shot classification accuracies than the CPT model, and the performance of FBCL with AMP is close to that of CPT + AMP.

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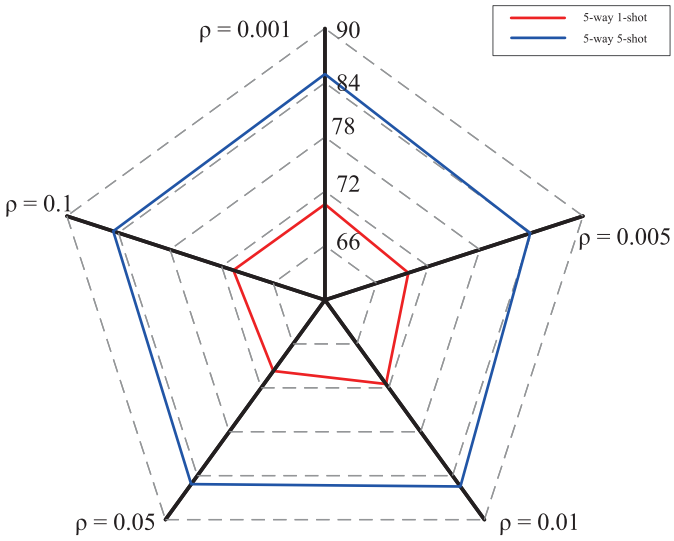


Fig. 6. FBS module classification accuracy under different values of parameter ρ.

Table I also reports the comparisons of each few-shot classification model on the AID dataset. It is clear that the proposed FBCL achieves the best results of 72.98% and 87.50% in both the cases, respectively.

As for UCM, compared with most methods, our FBCL achieves higher classification accuracies in both the five-way one-shot and five-way five-shot conditions. In IDLN, an attention-based feature calibration is proposed to optimize features based on label relations. In SGMNet, a novel scene-graph-matching-based framework is proposed. Both these methods introduce distributions of test sample to optimize the matching process, which makes the classification accuracies of these two methods higher than ours on the smaller dataset of UCM.

In summary, the proposed FBCL achieves excellent classification results on the three datasets, which performs a great improvement on the NWPU-RESISC45 and AID datasets. It can be concluded that the proposed FBCL can efficiently distinguish the foreground and background regions, which makes full utilization of the overall information of images to enhance the few-shot classification performance.

D. Parameter Analyses

In the proposed loss function, the threshold ρ in (8) is developed to ignore distinguishable foreground–background pairs, which has an effect on the classification result. In the experiments, the effect of ρ is analyzed. As shown in Fig. 6, classification accuracies under different values of parameter ρ are depicted, where the red line indicates the performance in the five-way one-shot case, and the blue line indicates the performance in the five-way five-shot case. It can be seen that our model yields the optimal classification results when ρ is equal to 0.01. Therefore, the threshold ρ is set to 0.01 in our model.

In the overall loss function, α is the scale factor to balance the proposed loss and the supervised contrastive loss, which also has an effect on the classification performance.

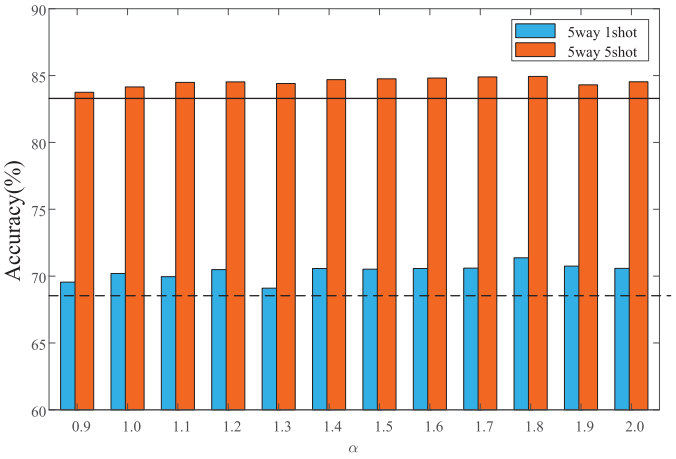


Fig. 7. FBS module classification accuracy under different values of parameter α.

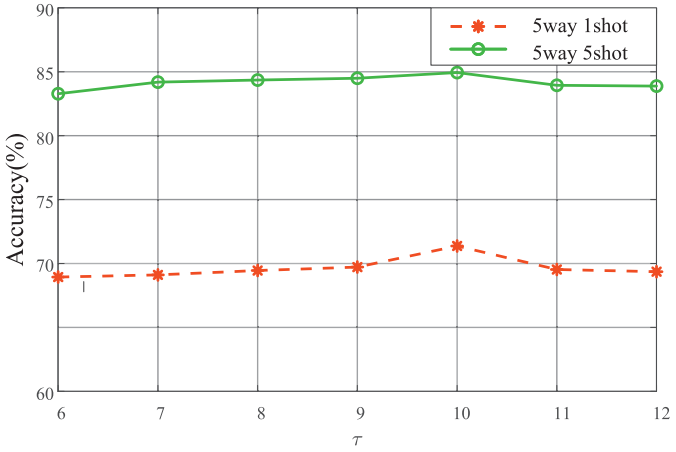


Fig. 8. FBS module classification accuracy under different values of parameter τ.

Fig. 7 depicts classification accuracies under different values of parameter α, where α is ranging from 0.9 to 2.0. It is observed that the change in classification accuracy is not very significant under the five-way five-shot case, and the highest accuracy is arrived when α is equal to 1.8. Under the five-way one-shot case, classification performance tends to increase first and then decrease, and the optimal accuracy is also obtained when α is equal to 1.8. Thus, the scale factor α is set to 1.8 in our model.

In addition, the temperature coefficient τ in supervised comparative loss is also analyzed. Fig. 8 depicts the accuracies of our model with different τ, where τ is ranging from 6 to 12. It can be seen that the classification accuracy increases first and then decreases with the change in τ, and the accuracy performs the highest with $\tau = 10$. Thus, the temperature coefficient τ is set to 10 in the proposed model.

E. Visual Analyses

1) Visual Analysis of Feature Heat Map: To more intuitively illustrate the validity of our model, class activation mapping (CAM) [58] is used to draw the feature heat maps. The visualization is performed by overlaying the heat maps with the original images. The redder the color, the greater

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TABLE I

RESULTS OF COMPARISON EXPERIMENTS ON THE NWPU-RESISC45, AID, AND UCM DATASETS UNDER FIVE-WAY. THE BEST RESULTS HAVE BEEN HIGHLIGHTED

Method	NWPU-RESISC45		AID		UCM	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
Relation network [36]	63.27 ± 0.81	78.53 ± 0.51	58.83 ± 0.62	70.02 ± 0.69	50.23 ± 0.92	63.48 ± 0.47
Matching network [17]	37.70 ± 0.65	47.29 ± 0.33	40.91 ± 0.86	64.18 ± 0.39	37.60 ± 0.88	52.81 ± 0.42
Prototypical network [35]	41.07 ± 0.45	62.78 ± 0.29	60.70 ± 1.02	74.83 ± 0.62	51.15 ± 0.72	63.40 ± 0.56
Meta-Baseline [49]	65.33 ± 0.69	81.71 ± 0.44	63.47 ± 0.90	77.33 ± 0.58	54.22 ± 0.76	67.27 ± 0.60
DLA-MatchNet [50]	68.80 ± 0.70	81.63 ± 0.46	57.21 ± 0.82	73.45 ± 0.61	53.76 ± 0.62	63.01 ± 0.51
SCL-MLNet [48]	62.21 ± 1.12	80.86 ± 0.76	59.46 ± 0.96	76.31 ± 0.68	51.37 ± 0.79	68.09 ± 0.92
TSC [51]	73.26 ± 0.15	84.62 ± 0.35	68.21 ± 0.83	81.72 ± 0.54	55.11 ± 0.68	69.20 ± 0.64
SPNet [52]	67.84 ± 0.87	83.94 ± 0.50	-	-	57.64 ± 0.73	73.52 ± 0.51
AO w/o trans [54]	73.50 ± 0.44	87.90 ± 0.25	69.51 ± 0.44	85.74 ± 0.25	-	-
IDLN [55]	75.25 ± 0.75	84.67 ± 0.23	68.65 ± 0.46	80.58 ± 0.27	60.09 ± 0.44	73.27 ± 0.25
SGMNet [56]	73.01 ± 0.77	84.52 ± 0.50	64.32 ± 0.79	77.98 ± 0.42	**64.17 ± 0.75**	**76.63 ± 0.59**
CPT [53]	73.50 ± 0.44	87.90 ± 0.23	69.51 ± 0.44	85.74 ± 0.25	-	-
FBCL(Ours)	**75.35 ± 0.82**	**88.90 ± 0.41**	**72.37 ± 0.70**	**86.91 ± 0.38**	58.45 ± 0.74	74.59 ± 0.59
CPT + AMP	**76.70 ± 0.44**	**89.87 ± 0.41**	72.67 ± 0.43	87.38 ± 0.23	-	-
FBCL(Ours) + AMP	76.41 ± 0.75	89.40 ± 0.51	**72.98 ± 0.79**	**87.50 ± 0.40**	-	-

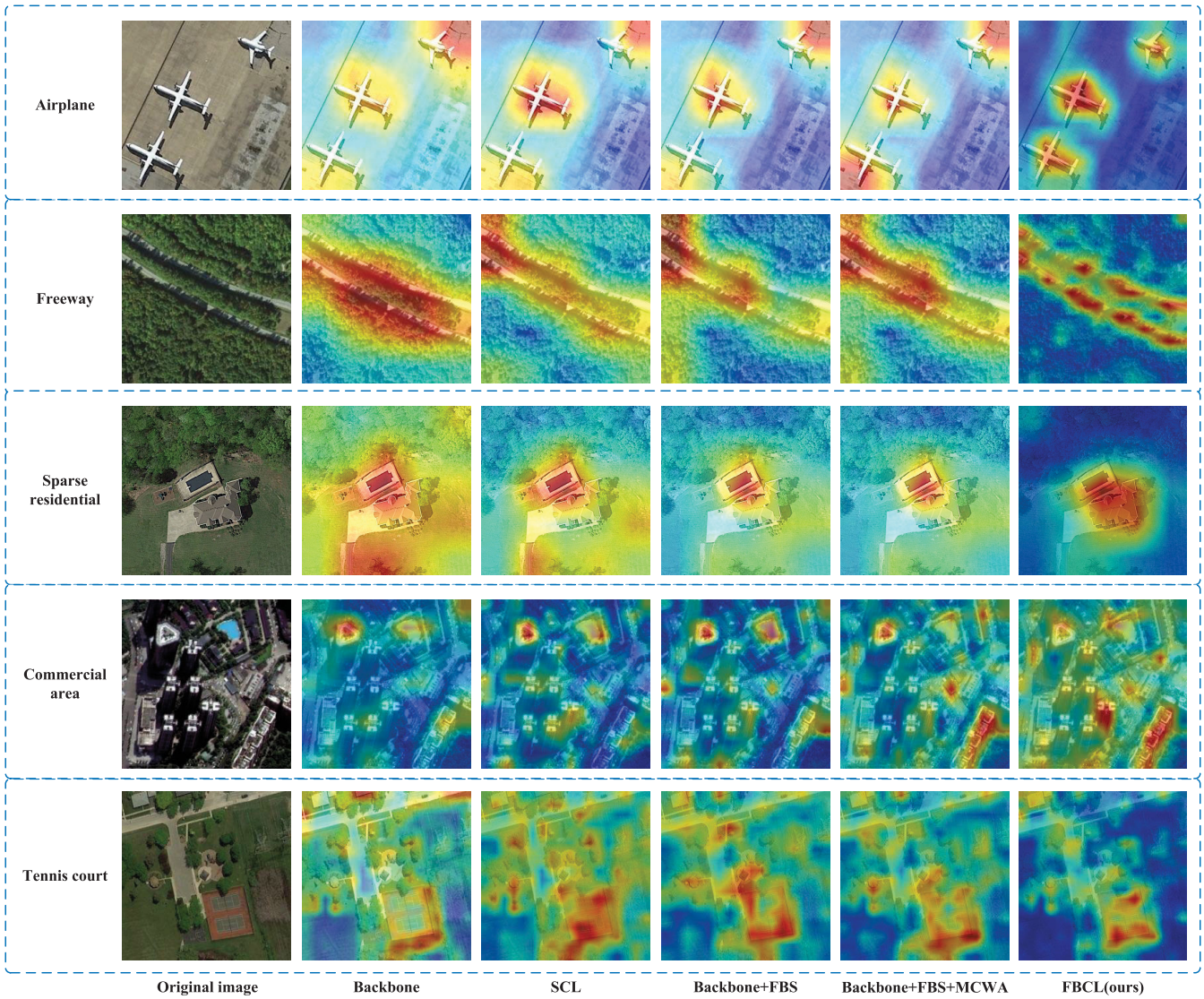


Fig. 9. CAM of partial samples on the NWPU-RESISC45 dataset.

the weight, indicating that the model pays more attention to the area. Thus, salient region highlighted by the heat map contributes much higher to the final classification results.

In Fig. 9, five samples are randomly selected from the three datasets and their CAM figures under the five conditions are shown. Backbone denotes training feature extractor using

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cross-entropy loss, SCL denotes using supervised contrastive loss to train the model, Backbone + FBS stands for the proposed FBS module, Backbone + FBS + MCWA means the channel weight allocator added to Backbone + FBS, and finally, FBCL is the overall proposed method, i.e., in the experiments, cosine distance classifier is used for all the five above conditions.

It is clearly seen from Fig. 9 that the heat map of the backbone can roughly identify the target area, but the recognition of the object is too coarse and part salient areas fall on the background. Compared with Backbone, SCL performs a much finer recognition of the object region. Based on the results of Backbone + FBS, it illustrates that our FBS module yields more detailed edges of objects compared with SCL based on mask, since the proposed FBS module can effectively distinguish the foreground and background. Moreover, our proposed framework performs the optimal heat map, where the object region can be effectively highlighted. As for airplane, Backbone + FBS + MCWA yields much better heat map than the compared approaches, where FBS and SCL can only focus on two of the airplanes. It can demonstrate that MCWA further improves the advantage of RS image information. Compared with Backbone + FBS + MCWA, the attention areas of FBCL are more accurate, which proves that MFF enables the model to pay more accurate attention on the target regions.

2) Visual Analysis of Feature Distribution: To verify the efficacy of feature extractor in FBCL, the T-SNE algorithm is used to reflect the distribution of sample features. To better prove the effectiveness of our method, all the categories of the test set are applied for visual analysis, where 200 samples per class are randomly selected from the test set of NWPU-RESISC45. Visual results of feature distributions are depicted in Fig. 10, where different colored points stand for different categories. The feature distributions of the prototype network, SCL model, and our FBCL are depicted in Fig. 10(a)–(c), respectively. Compared with other two approaches, feature distribution of our proposed model is more concentrated on the category prototypes. It is demonstrated that the proposed FBCL can effectively distinguish the foreground and background regions so as to extract more discriminative features.

F. Ablation Studies

1) Ablation Studies of Different Modules: This section conducts ablation experiments between different modules to verify the effectiveness of each module. Specifically, the following seven conditions are considered for comparisons.

1) *Backbone:* ResNet-12 is used as the feature extractor and trained with the cross-entropy loss.
2) *Backbone + FBS:* The FBS module is added to the backbone, which can enhance the discrimination between the foreground features and background features.
3) *Backbone + SCWA:* The backbone without the feature separation module is constructed, and single-feature channel weight allocation (SCWA) is added.
4) *Backbone + FBS + MFF:* The FBS module and the MFF module are added to the backbone.
5) *Backbone + FBS + CS:* The FBS module is added to the backbone, and features are channel stitched (CS) for classification.
6) *Backbone + FBS + MCWA:* The FBS module is added to the backbone, and multifeature channel weight allocation is applied to further optimize features.
7) *FBCL (Backbone + FBS + MFF + MCWA):* Our proposed framework is constructed, where the MFF module and the FBS module are added to the backbone, and multifeature channel weight allocation is added to further optimize features.

TABLE II
ABLATION STUDIES BETWEEN THE DIFFERENT MODULES ON THE NWPU-RESISC45 DATASET

Method	5-way 1-shot	5-way 5-shot
Backbone	64.33 ± 0.82	81.89 ± 0.57
Backbone+FBS	71.48 ± 0.81	85.48 ± 0.53
Backbone+SCWA	71.66 ± 0.85	86.83 ± 0.45
Backbone+FBS+CS	70.33 ± 0.84	87.08 ± 0.42
Backbone+FBS+MFF	71.93 ± 0.78	88.11 ± 0.42
Backbone+FBS+MCWA	73.50 ± 0.84	87.91 ± 0.43
FBCL (Ours)	**75.35 ± 0.82**	**88.90 ± 0.41**

TABLE III
ABLATION STUDIES BETWEEN THE DIFFERENT MODULES ON THE UCM DATASET

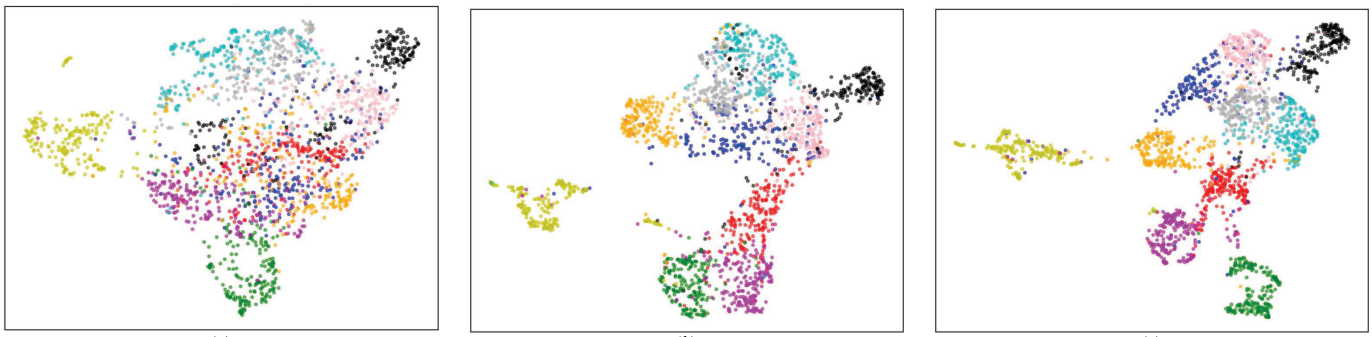
Method	5-way 1-shot	5-way 5-shot
Backbone	43.62 ± 0.62	58.51 ± 0.56
Backbone+FBS	54.44 ± 0.81	68.24 ± 0.50
Backbone+SCWA	54.77 ± 0.78	68.53 ± 0.51
Backbone+FBS+CS	54.52 ± 0.74	69.64 ± 0.52
Backbone+FBS+MFF	55.04 ± 0.78	72.41 ± 0.55
Backbone+FBS+MCWA	55.38 ± 0.71	69.77 ± 0.50
FBCL (Ours)	**58.45 ± 0.74**	**74.59 ± 0.59**

The results of ablation studies on three datasets are reported in Tables II–IV, respectively. Comparing Backbone + FBS with Backbone, it can be seen that adding the FBS module is able to improve the few-shot classification results for both the five-way one-shot and five-way five-shot cases. Compared with Backbone + FBS, Backbone + FBS + MFF achieves better results on all the three datasets. Similarly, the accuracies of Backbone + SCWA are higher than Backbone in all the cases, which demonstrates that it is reasonable to introduce a channel weight allocation module to our deep network. Finally, classification accuracies of our FBCL are higher than the compared six methods in all the cases.

In summary, the FBS module proposed in our framework can improve the ability to distinguish between the foreground and background regions of RS images, the MFF module enables the model to pay more accurate attention on the target regions, and the multifeature channel weight allocation module can reasonably use the feature information, which all have significant contributions to enhance the few-shot classification performance.

2) Ablation Studies of Different Losses: In the FBS module, there are three loss functions. To demonstrate the contribution of different losses, an ablation experiment is presented as shown in Table V.

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(a) (b) (c)

Fig. 10. Feature visualization by T-SNE. (a) Prototypical network, (b) SCL, and (c) FBCL (ours).

TABLE IV

ABLATION STUDIES BETWEEN THE DIFFERENT MODULES ON THE AID DATASET

Method	5-way 1-shot	5-way 5-shot
Backbone	60.41 ± 0.74	77.74 ± 0.51
Backbone+FBS	70.51 ± 0.81	83.82 ± 0.47
Backbone+SCWA	70.24 ± 0.76	83.42 ± 0.46
Backbone+FBS+CS	70.46 ± 0.78	85.46 ± 0.46
Backbone+FBS+MFF	71.68 ± 0.79	86.35 ± 0.45
Backbone+FBS+MCWA	71.22 ± 0.77	85.94 ± 0.44
FBCL (Ours)	**72.37 ± 0.70**	**86.91 ± 0.38**

TABLE V

ABLATION STUDIES OF DIFFERENT LOSS FUNCTIONS ON THE NWPU-RESISC45 DATASET

$\mathcal{L}^{sup}$	$\mathcal{L}^{f-b}$	$\mathcal{L}^{f-f}$	NWPU-RESISC45	
			1-shot	5-shot
✓			66.95 ± 0.82	82.45 ± 0.57
✓		✓	68.71 ± 0.83	84.43 ± 0.49
✓	✓		69.53 ± 0.84	84.56 ± 0.48
✓	✓	✓	**71.48 ± 0.81**	**85.48 ± 0.53**

It is observed from the above table that the overall loss function combining supervised contrast loss $\mathcal{L}^{\text{sup}}$, foreground–foreground loss $\mathcal{L}^{f-f}$, and foreground–background loss $\mathcal{L}^{f-b}$ produces the optimal classification results both in the five-way one-shot and five-way five-shot cases. Only using the supervised contrast loss $\mathcal{L}^{\text{sup}}$, the accuracy performs 4.53% and 3.03% lower than combining three losses under both the conditions, respectively. Comparing foreground–foreground loss $\mathcal{L}^{f-f}$ and foreground–background loss $\mathcal{L}^{f-b}$, it indicates that foreground–background loss $\mathcal{L}^{f-b}$ has higher contribution than foreground–foreground loss $\mathcal{L}^{f-f}$ for few-shot classification. In addition, it is illustrated that the FBS module has important influence for improving few-shot scene classification.

3) Ablation Studies With Different Shots: To further prove the validity of our method, we have compared the classification accuracy under different shots on the NWPU-RESISC45 dataset between SCL, meta-baseline, TSC, IDLN, and our FBCL in Fig. 11, where the shot number is changed from 1 to 10.

It can be observed from the results that our proposed FBCL yields superior accuracies with different shot numbers than other compared approaches, which further prove the validity of the proposed model.

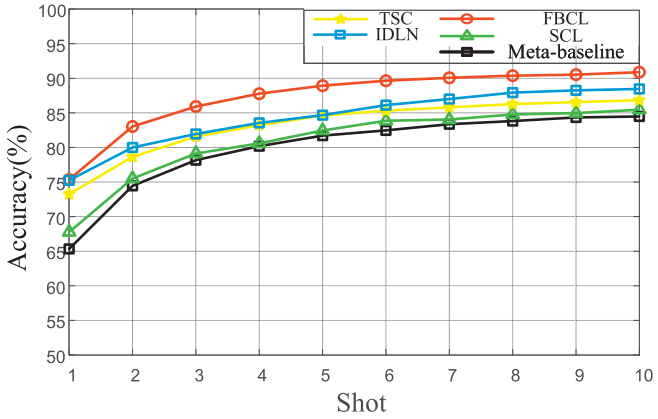


Fig. 11. Classification accuracy under different shots on the NWPU-RESISC45 dataset.

V. CONCLUSION

In this article, an FBCL is proposed for few-shot scene classification of RS images. The experimental results on three datasets demonstrate that the proposed model outperforms other related approaches. The MFF module enables the model to pay more accurate attention on the target regions, and the FBS module is optimized with SCL, which is proved to effectively distinguish the foreground and background regions of RS images. Moreover, the multifeature channel weight allocation is able to balance features of different dimensions, which can make full use of the separated features to improve the performance.

Since SCL is applied in our framework, it requires label information for feature learning. In future work, we prefer to introduce the self-SCL that can effectively use the large number of unlabeled samples.

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